

Problem 16.1

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A^T A \hat{x} = A^T b$$

$$b = \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$\leadsto \begin{bmatrix} 1 & 2/3 \\ 0 & 32/6 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 35 \\ 42 \end{bmatrix}$$

$$\Rightarrow D = \frac{42 \times 3}{32} = \frac{126}{8} = 15.75$$

$$C = \frac{35 - 2 \times 15.75}{3} = 9$$

Problem 16.2

$$P = A \begin{bmatrix} 9 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 9 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix}$$

$$e = \begin{bmatrix} 7 \\ 7 \\ 21 \end{bmatrix} - \begin{bmatrix} 5 \\ 13 \\ 17 \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ +4 \end{bmatrix}$$

because $e \perp P$.

Problem 16.3

If A is the projection vector then $I - A$ is the projection of errors

$$\begin{bmatrix} 0 \\ -1 \\ -1 \end{bmatrix}$$

$$P = A$$

$b \perp$ to Column Space of A

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix}$$

$$A^T A \hat{x} = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} = A^T \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -6 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Subject:

Year:

Month:

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Problem 16.4

$$A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 6 & 13 \\ 1 & 17 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 5 \\ 4 & 2 \end{bmatrix}$$

$$-7C = 63$$

because b is in $C(A)$

$$C = 9$$

$$D = 4$$

Problem 16.5

Left Null Space,

Column Space,

~~Null Space~~ $C(A^T)$,

$$N(A) = \{0\}$$

Problem 16.6

$$C = 1, D = -1$$